

Parser generator

When using parser generator, there is an emphasis on decomposition and reusability. By decomposition, I mean that the parsing process is split into two: the *lexing* stage (a.k.a. character grouping) and the *parsing* stage; a more detailed pipeline can be found in Phan Thi Tuoi's compilation [1]. By reusability, I mean that we interpret the parsing process in terms of intermediate forms (e.g. the abstract syntax tree), so that the parsing process is simply a multiple of transformation steps, and each step can be branching checkpoint to produce other kinds of outputs.

Hand-written parser

Fused parser

PPX-based parser

where PPX stands for *preprocessing extension*.

From another dimension, it's important to grasp the *parsing model*. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliquam quaerat voluptatem. Ut enim aequaeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

Parsing using regular expression

Parsing using parser combinators

Parsing in the bkhack shell implementation

Reference

- [1] P. T. Tươi, *Giáo trình Trình biên dịch*. Nhà xuất bản Đại học Quốc gia TP.HCM.